

Name: _____

FRACTIONS ALL OPERATIONS, (MULTIPLYING FRACTIONS)

Date: _____

LO: I can multiply fractions and mixed numbers by multiplying numerators and denominators.

Multiplying fractions

To multiply fractions, multiply the numerators together and multiply the denominators together. Simplify if possible.

Tip: You can cross-cancel before multiplying to keep numbers small.

Quick examples

●	$2\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$	multiply tops, multiply bottoms
●	$\frac{3}{4} \times \frac{2}{9} = \frac{1}{6}$	cross-cancel 3 and 9
●	$\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$	unit fractions
●	$\frac{2}{3} \times \frac{4}{5} = \frac{6}{8}$	added instead of multiplied
●	$\frac{3}{4} \times \frac{2}{9} = \frac{6}{36}$	correct but not simplified

Key idea

Multiply the numerators together and the denominators together. Simplify the result.

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Top times top, bottom times bottom

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - simple multiplication

Calculate $2\frac{2}{5} \times \frac{3}{4}$

Multiply numerators: $2 \times 3 = 6$

Multiply denominators: $5 \times 4 = 20$

$$\frac{6}{20} = \frac{3}{10}$$

$$= \frac{3}{10}$$

Always simplify your final answer.

Example 2 - cross-cancelling

Calculate $3\frac{8}{9} \times \frac{4}{9}$

Cancel 3 and 9 (common factor 3):

$$\frac{1}{8} \times \frac{4}{9}$$

Cancel 4 and 8 (common factor 4):

$$\frac{1}{2} \times \frac{1}{9}$$

$$= \frac{1}{6}$$

$$= \frac{1}{6}$$

Cross-cancelling makes the multiplication simpler.

Example 3 - multiplying by a whole number

Calculate $3\frac{7}{7} \times 5$

Write 5 as $\frac{5}{1}$

$$\frac{3}{7} \times \frac{5}{1} = \frac{15}{7}$$

$$= 2\frac{1}{7}$$

$$= 2\frac{1}{7}$$

Write whole numbers as fractions with denominator 1.

Example 4 - mixed numbers

Calculate $11\frac{1}{3} \times 2\frac{1}{4}$

Convert: $\frac{4}{3} \times \frac{9}{4}$

Cancel 4s: $\frac{1}{3} \times \frac{9}{1}$

Cancel 3 and 9: $\frac{1}{1} \times \frac{3}{1} = 3$

$$= 3$$

Convert to improper fractions, then cross-cancel.

YOUR TURN – PRACTICE (deliberate practice)

1. Unit fractions

Multiply these unit fractions.

- a) $\frac{1}{2} \times \frac{1}{3}$
- b) $\frac{1}{4} \times \frac{1}{5}$
- c) $\frac{1}{3} \times \frac{1}{7}$
- d) $\frac{1}{6} \times \frac{1}{4}$

2. Non-unit fractions

Multiply and simplify.

- a) $\frac{2}{3} \times \frac{3}{5}$
- b) $\frac{3}{4} \times \frac{2}{7}$
- c) $\frac{4}{5} \times \frac{5}{8}$
- d) $\frac{5}{6} \times \frac{3}{10}$

3. Multiplying by whole numbers

Find each product.

- a) $\frac{2}{5} \times 4$
- b) $\frac{3}{8} \times 6$
- c) $\frac{5}{6} \times 3$
- d) $\frac{7}{10} \times 5$

4. Mixed numbers

Convert to improper fractions first.

- a) $1\frac{1}{2} \times \frac{2}{3}$
- b) $2\frac{1}{4} \times \frac{4}{9}$
- c) $1\frac{2}{5} \times 2\frac{1}{2}$
- d) $3\frac{1}{3} \times \frac{3}{5}$

5. Mixed multiplication

Calculate and simplify.

- a) $\frac{4}{9} \times \frac{3}{8}$
- b) $\frac{5}{7} \times 14$
- c) $2\frac{1}{3} \times 1\frac{2}{7}$
- d) $\frac{6}{11} \times 1\frac{1}{12}$
- e) $1\frac{3}{4} \times 2\frac{2}{7}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$ ☐

Correct answer: _____

Reason: _____

b) $\frac{2}{3} \times \frac{4}{5} = \frac{6}{8}$ ☐

Correct answer: _____

Reason: _____

c) You can cross-cancel before multiplying ☐

Correct answer: _____

Reason: _____

d) Multiplying fractions always gives a bigger number ☐

Correct answer: _____

Reason: _____

e) $1\frac{1}{2} \times \frac{2}{3} = 1$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangular garden is $4\frac{1}{2}$ m long and $2\frac{2}{3}$ m wide. Find the area. If fencing costs £8 per metre, how much does it cost to fence the perimeter?

Expression: _____

Simplified: _____

b) In a class of 30 pupils, $\frac{2}{5}$ are boys. Of the boys, $\frac{3}{4}$ play football. What fraction of the whole class are boys who play football? How many pupils is this?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

